

Inhibiting NINJ1-dependent plasma membrane rupture protects against inflammasome-induced blood coagulation and inflammation

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eLife Assessment

The authors aim to elucidate the mechanism by which pyroptosis (through the formation of Gasdermin D (GSDMD) pores in the plasma membrane) contributes to increased release of procoagulant Tissue Factor-containing microvesicles. The data offers **solid** mechanistic insights as to the interplay between pyroptosis and microvesicle release with NINJ1. The study provides **useful** insights into the potential of targeting Ninj1 as a therapeutic strategy.

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Abstract

Systemic blood coagulation accompanies inflammation during severe infection like sepsis and COVID. We've previously established a link between pyroptosis, a vital defense mechanism against infection, and coagulopathy. During pyroptosis, the formation of gasdermin-D (GSDMD) pores on the plasma membrane leads to the release of tissue factor (TF)-positive microvesicles (MVs) that are procoagulant. Mice lacking GSDMD release fewer TF MVs. However, the specific mechanisms leading from activation of GSDMD to MV release remain unclear. Plasma membrane rupture (PMR) in pyroptosis was recently reported to be actively mediated by the transmembrane protein Ninjurin-1 (NINJ1). Here we show that NINJ1 promotes procoagulant MV release during pyroptosis. Haploinsufficiency or glycine inhibition of NINJ1 limited the release of procoagulant MVs and inflammatory cytokines and partially protected against blood coagulation and lethality triggered by bacterial flagellin. Our findings suggest a crucial role for NINJ1-dependent PMR in inflammasome-induced blood coagulation and inflammation.

Introduction

Inflammasomes are complex intracellular sensors that detect pathogens by sensing pathogen-associated molecular patterns (PAMPs) (1, 2). The activation of inflammasome and subsequent pyroptosis is critical in host defense against pathogens (3–5). Pyroptosis exposes intracellular pathogens through plasma membrane rupture (PMR) (6). However, overactive inflammasomes, common in conditions like sepsis and COVID (7–11), can trigger systemic blood coagulation and tissue and organ dysfunction (12–16).

Tissue factor (TF) is a membrane protein and the primary initiator of the coagulation cascade, essential for hemostasis to stop bleeding upon injury (17, 18). Increased TF activity is associated with consumption coagulopathy linked to sepsis and COVID (19–22). Our previous study has shown that TF-positive MVs released by pyroptotic macrophages trigger systemic coagulation and lethality in experimental sepsis (23). However, the molecular mechanisms by which pyroptosis releases procoagulant MVs remain elusive. Pyroptotic cells undergo PMR at the late stage when the cell membrane is beyond repair, where the direct blebbing and pinching of the plasma membrane releases MVs to remove membrane damaged with GSDMD pores (24–28). PMR could accelerate MV release as the entire plasma membrane is subject to dissolve as MVs. Kayagaki et al. recently reported that NINJ1 oligomerization activates PMR, a process formerly thought to be passive, during lytic cell death like necrosis and pyroptosis (29). The role of NINJ1 in releasing procoagulant MVs and initiating systemic coagulation after inflammasome activation remains unknown.

In this study, we show that *Ninj1* haploinsufficiency inhibits the release of TF-positive MVs and provides protection against systemic coagulation. Reduced NINJ1 also limits the release of inflammatory cytokines. We find that a single dose of glycine injection has similar protective effects as they prevent PMR. Our findings underscore the essential role of NINJ1-dependent PMR in inflammasome-induced blood coagulation, thus expanding our understanding of the dual role that inflammasomes play in host defense.

Results and discussion

***Ninj1* haploinsufficiency protects against flagellin-induced blood coagulation**

We aimed to understand the role of NINJ1 on inflammasome-induced blood coagulation, using heterozygous *Ninj1*^{+/-} mice with decreased NINJ1 protein abundance (Fig S1). This approach was justified by our observation that total NINJ1 deficiency could lead to partial embryonic lethality and the reported occurrence of hydrocephalus in surviving mice (30).

To induce blood coagulation and lethality, we applied our established protocol the involves injection of mice with purified bacterial flagellin, fused with the cytosolic translocation domain of anthrax lethal factor (LFn). In the presence of anthrax protein protective agent (PA), this flagellin fusion protein (LFn-Fla) can be transported into the cytoplasm (4, 31). Flagellin is a PAMP known to activate the NAIP/NLRC4 inflammasome and trigger pyroptosis (3, 4, 32, 33). We confirmed robust inflammasome activation and pyroptosis in mouse primary bone marrow-derived macrophages (BMDMs) upon stimulating with purified Fla (LFn-Fla plus PA) (Fig S2A-B).

Our earlier work has shown that inflammasome-induced systemic coagulation is marked by prolonged prothrombin time (PT) and increased plasma thrombin-antithrombin (TAT) (23 [↗](#)), frequently observed in patients with disseminated intravascular coagulation (DIC) and COVID (12 [↗](#)–16 [↗](#), 34 [↗](#)). As expected, intravenous injection of Fla activated systemic coagulation in wild-type *Ninj1*^{+/+} mice, demonstrated by a significant increase of PT and a rise in plasma TAT (Fig 1A–B [↗](#)). Notably, when challenged with Fla, heterozygous *Ninj1*^{+/-} mice showed lower PT and almost complete suppression of the elevation in plasma TAT compared with their wild-type littermates (Fig 1A–B [↗](#)). Our results suggest that a single copy deletion of *Ninj1* is adequate for significant inhibition of inflammasome-induced blood coagulation.

We also found that *Ninj1* haploinsufficiency hindered the release of cytokines, as shown by decreased plasma concentrations of cytokines like IL-1 β , TNF α , and IL-6 (Fig 1C–E [↗](#)). The decrease in cytokine concentrations could be attributed to limited release of danger-associated molecular patterns (DAMPs) due to less effective PMR in *Ninj1*^{+/-} mice. It's worth noting that GSDMD facilitates IL-1 β and NINJ1 deficiency does not affect pyroptosis-induced IL-1 β release into cell culture supernatant from BMDMs (ref). Given that *Ninj1* haploinsufficiency did not change GSDMD expression (Fig 1S [↗](#)), there might be unrecognized mechanisms in pyroptosis-induced IL-1 β *in vivo*.

While our earlier research suggested that cell death, not the release of inflammatory cytokines like IL-1 β , is crucial for the over-activation of blood coagulation triggered by inflammasomes (23 [↗](#)), these cytokines are vital for maintaining inflammation. Our findings indicate their release is facilitated by NINJ1.

Fibrin deposition and lethality resulting from inflammasome activation depend on NINJ1

Fibrin deposition in tissues can restrict blood supply, potentially resulting in tissue and organ damage and even death (12 [↗](#)). Following the Fla challenge, we noted fibrin deposition in the lungs of wild-type mice through immunostaining with a fibrin-specific antibody 59D8 (ref) (Fig 1F [↗](#)), which was consistent with H&E staining (Fig 3S [↗](#)). However, no signs of fibrin deposition were observed in lung sections of *Ninj1*^{+/-} mice, likely due to the technical limitations of immunostaining on thin (5- μ m) tissue sections, compounded by the potential for uneven fibrin distribution. By using immunoblot to examine a homogenized whole lobe of the lung or liver, we detected fibrin deposition in the liver and lungs of *Ninj1*^{+/-} mice, but to a much less extent compared with their *Ninj1*^{+/+} littermates (Fig 1G [↗](#)). Furthermore, *Ninj1* haploinsufficiency partially rescued flagellin-induced lethality (Fig 1H [↗](#)). These findings underscore the significant role of NINJ1 protein in inflammasome-triggered fibrin deposition in tissues and lethality.

NINJ1 plays a critical role in systemic coagulation in response to *E. coli* infection

Systemic blood coagulation is a common complication of pathogen-triggered sepsis (35 [↗](#), 36 [↗](#)). To investigate the role of NINJ1 in bacterial infection-induced blood coagulation, *Ninj1*^{+/-} mice were infected with cultured *E. coli* at a dose of 2×10^8 pfu per mouse. Consistently with the findings from the Fla challenge, *E. coli* infection-induced blood coagulation was limited by *Ninj1* haploinsufficiency. As shown in Fig 2A–B [↗](#), *Ninj1*^{+/-} mice exhibited a lower PT and decreased plasma TAT following *E. coli* infection. Furthermore, *Ninj1*^{+/-} mice displayed alleviated cytokine release in response to *E. coli* infection (Fig 2C–E [↗](#)). Collectively, these data demonstrate that reduced expression of NINJ1 confers protection against *E. coli* infection-induced blood coagulation. In addition, the alleviation of cytokine release in *Ninj1*^{+/-} mice supports the role of NINJ1 in modulating the inflammatory response to *E. coli* infection. These findings highlight a potential role of NINJ1 in the pathogenesis of bacterial infection, including blood coagulation and the cytokine storm.

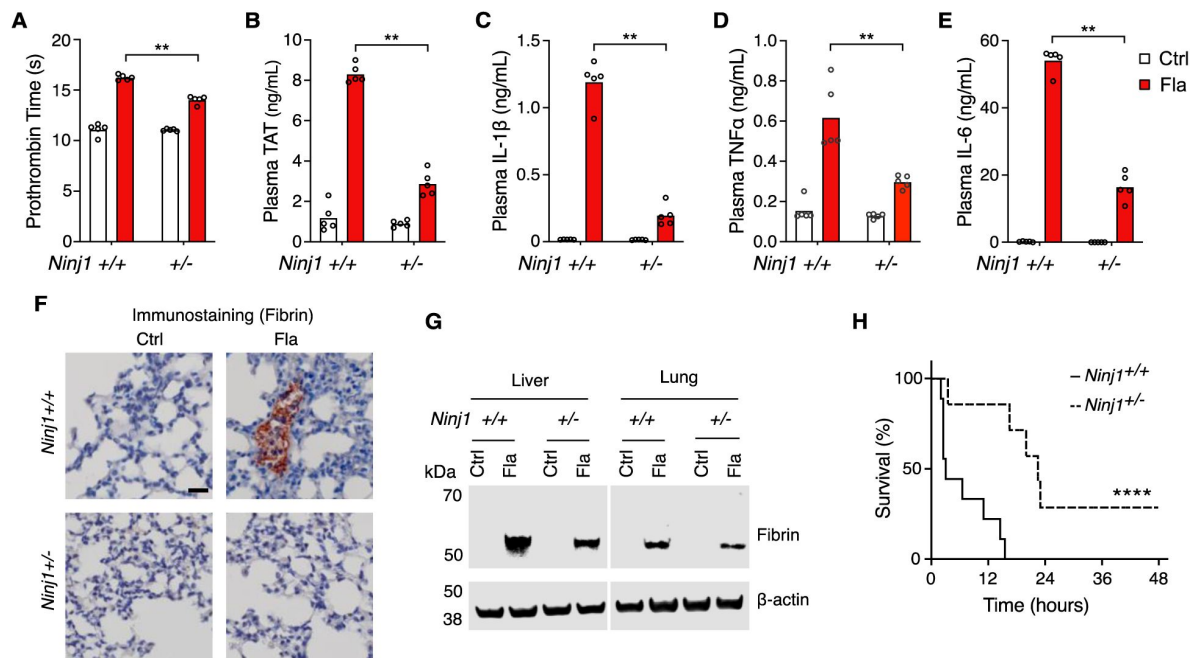


Figure 1.

NINJ1 is critical for flagellin-induced systemic coagulation, inflammation, and lethality.

(A–E) Mice were injected intravenously with Ctrl (saline) or Fla (500 ng LFn-Fla plus 3 μ g PA). Blood was collected 90 minutes after Ctrl or Fla injection. Prothrombin time (A), plasma TAT (B), and plasma cytokines (C–E) were measured. Circles represent individual mouse, with bars donating mean. ** $p < 0.01$ (two-way ANOVA with Holm-Sidak multiple comparisons).

(F–G) Mice were injected intravenously with Ctrl or Fla. After 90 minutes, mice were euthanized and perfused with PBS, and tissues were isolated. (F) Lung sections were stained with the anti-fibrin monoclonal antibody (59D8). Scale bar denotes 20 μ m. Data are representative of 3 independent experiments (biological replicates). (G) Fibrin in the liver and lungs was detected by immunoblot with the anti-fibrin monoclonal antibody (59D8). Data are representative of 3 independent experiments (biological replicates).

(H) Mice were injected intravenously with a lethal dose of Fla (2.5 μ g LFn-Fla plus 6 μ g PA). Kaplan-Meier survival plots for mice challenged with Fla are shown. $n = 7-9$. **** $p < 0.0001$ versus WT (log rank test [Mantel-Cox]).

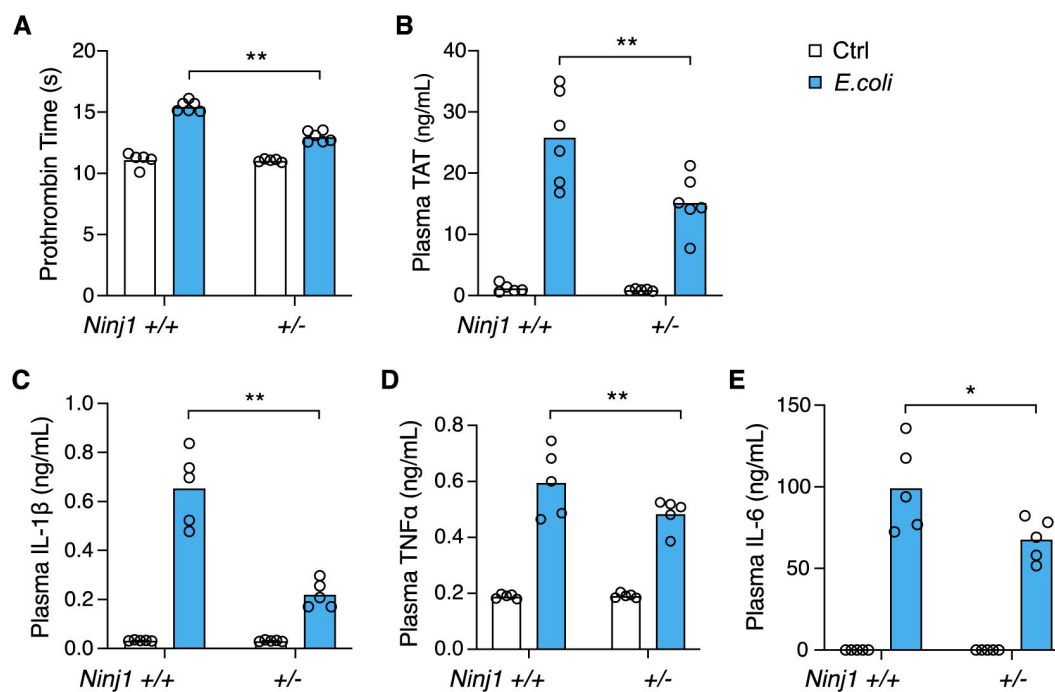


Figure 2.

***E. coli* infection-induced blood coagulation is limited in *Ninj1*^{+/-} mice.**

(A–E) Mice were injected intravenously with Ctrl (saline) or *E. coli* (2 x 10⁸ cfu per mouse). Blood was collected 6 hours afterwards. Prothrombin time (A), plasma TAT (B), and plasma cytokines (C–E) were measured. Circles represent individual mouse, with bars donating mean. * p < 0.05, ** p < 0.01 (two-way ANOVA with Holm-Sidak multiple comparisons).

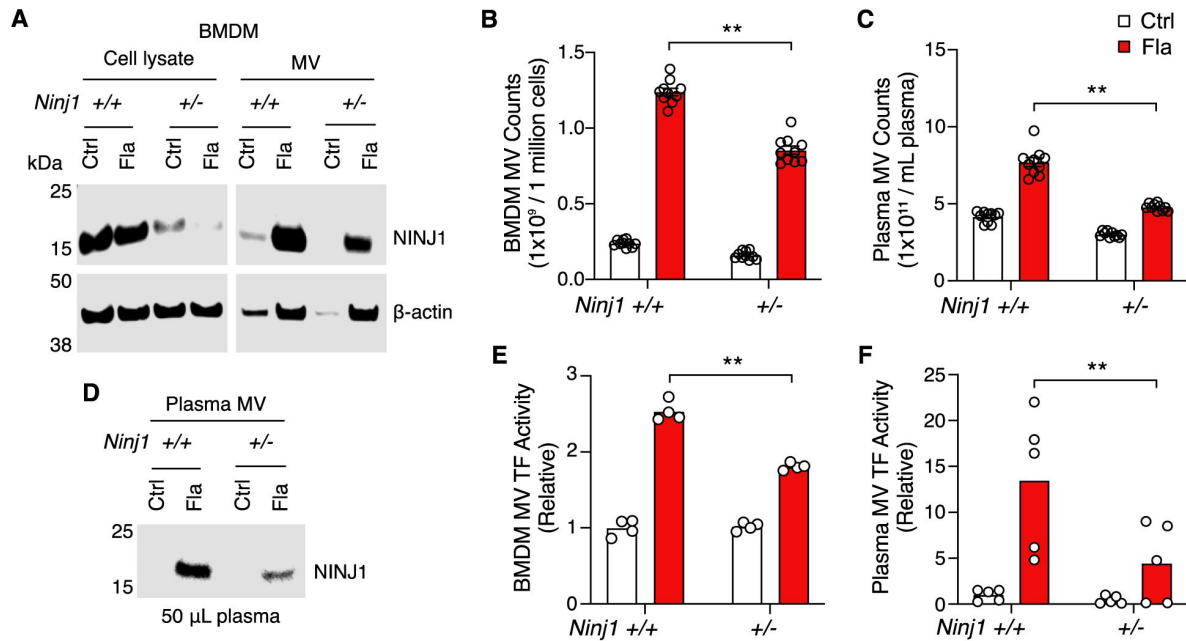


Figure 3.

PMR promotes the release of procoagulant MVs.

(A-B, E) BMDMs were incubated with Ctrl (PBS) or Fla (1 μ g/mL LFn-Fla plus 1 μ g/mL PA). Cell culture supernatant and MVs were collected after 90 minutes of incubation. (A) NINJ1 in the cell lysate and MVs was detected by immunoblot. (B) BMDM MVs were counted with Nanosight. (E) BMDM MV TF activity. Circles represent individual mouse, with bars donating mean. ** $p < 0.01$ (two-way ANOVA with Holm-Sidak multiple comparisons).

(C, D, F) Mice were injected intravenously with Ctrl (saline) or Fla (500 ng LFn-Fla plus 3 μ g PA). Blood was collected 90 minutes after Ctrl or Fla injection. (C) Plasma MVs were counted with NanoSight. (D) NINJ1 in plasma MVs was detected with equal plasma volumes by immunoblot. (F) Plasma MV TF activity. Circles represent individual mouse, with bars donating mean. ** $p < 0.01$ (two-way ANOVA with Holm-Sidak multiple comparisons).

NINJ1-dependent PMR promotes procoagulant MV release

In our previous work, we established that macrophage pyroptosis promotes the release of procoagulant MVs (23). Yet, the cellular processes linking pyroptosis to MV release remain to be defined. We hypothesize that PMR during pyroptosis could be the molecular event responsible for the release of these MVs. To test this hypothesis, we used BMDMs isolated from *Ninj1*^{+/+} and *Ninj1*^{+/-} littermates. We observed a significant reduction in PMR, indicated by the decreased lactate dehydrogenase (LDH) in culture medium, in *Ninj1*^{+/-} cells (Fig S3A), while cell death, measured by ATP levels, remained unaffected (Fig S3B). This is consistent with previous studies suggesting that NINJ1 mediates PMR but not cell death (29, 37).

Notably, we detected NINJ1 protein in MVs released from *Ninj1*^{+/+} BMDMs (Fig 3A). The protein level was decreased in MVs derived from *Ninj1*^{+/-} BMDMs (Fig 3A), likely due to reduced MV release and reduced NINJ1 protein in *Ninj1*^{+/-} cells (Fig 3B). Consistently, *in vivo*, NINJ1 was detected in plasma MVs from *Ninj1*^{+/+} mice exposed to Fla (Fig 3D). *Ninj1*^{+/-} mice also had decreased amount of NINJ1 from MVs, likely due to fewer plasma MVs (Fig 3C). The reduced quantity of MVs from both plasma and BMDMs in *Ninj1*^{+/-} mice correlated with lower TF activity (Fig 3E and 3F). Our data illustrate that a decline in NINJ1 expression significantly reduces PMR and MV release. These findings provide compelling evidence supporting a critical role of NINJ1-dependent PMR in procoagulant MV release. Furthermore, MV shedding removes NINJ1 from the plasma membrane, hinting at a possible role of NINJ1 in cell repair.

Glycine inhibition of NINJ1 alleviates pyroptosis-induced blood coagulation

It has long been recognized that glycine buffering protects against PMR during cell death, such as pyroptosis and necrosis (38, 39). The exact mechanisms underlying the glycine protective effects remained unclear until Borges et al. revealed that glycine inhibits NINJ1 oligomerization and shields cells from rupture (37). Yet, whether glycine buffering would block the release of PMR-induced TF MVs or limit pyroptosis-induced coagulation *in vivo* is unknown. To complement our genetic approach, we injected a glycine solution into wild-type mice, prior to Fla injection, aiming to inhibit NINJ1.

Intriguingly, following the Fla challenge, glycine injection led to a notable decrease in PT and plasma TAT (Fig 4A and 4B), demonstrating significant protection against flagellin-induced blood coagulation. Consistently, mice treated with glycine exhibited lower plasma MV TF activity post the Fla challenge (Fig 4C). *In vitro*, glycine buffering on isolated BMDMs protected against PMR, as denoted by a reduction LDH release (Fig S4A), and decreased MV TF activity (Fig S4B). However, we observed a discrepancy in MV counts between the *in vitro* and *in vivo* studies. Contrary to isolated BMDMs (Fig S4C), glycine-treated mice had higher MV counts irrespective of the Fla challenge (Fig 4D). We speculate that glycine might enhance MV release from cell types other than macrophages, independently of inflammasome activation. Lastly, glycine administration reduced the release of proinflammatory cytokines IL-1b and TNFa (consistent with previous research (40)), but not IL-6 (Fig 4E-G). In conclusion, the cytoprotective effect of glycine buffering resulted in improved outcomes in inflammasome-induced blood coagulation and inflammation.

In summary, our study employed both genetic and pharmacological approaches to inhibit NINJ1-dependent PMR. Our findings suggest that PMR serves as a pivotal molecular event responsible for the release of procoagulant MVs and the induction of systemic coagulation and inflammation in an experimental model of sepsis. This novel mechanism sheds light on the potential for developing NINJ1 inhibition as a promising therapeutic strategy for alleviating acute inflammation and coagulation-associated organs and tissue and organ damage.

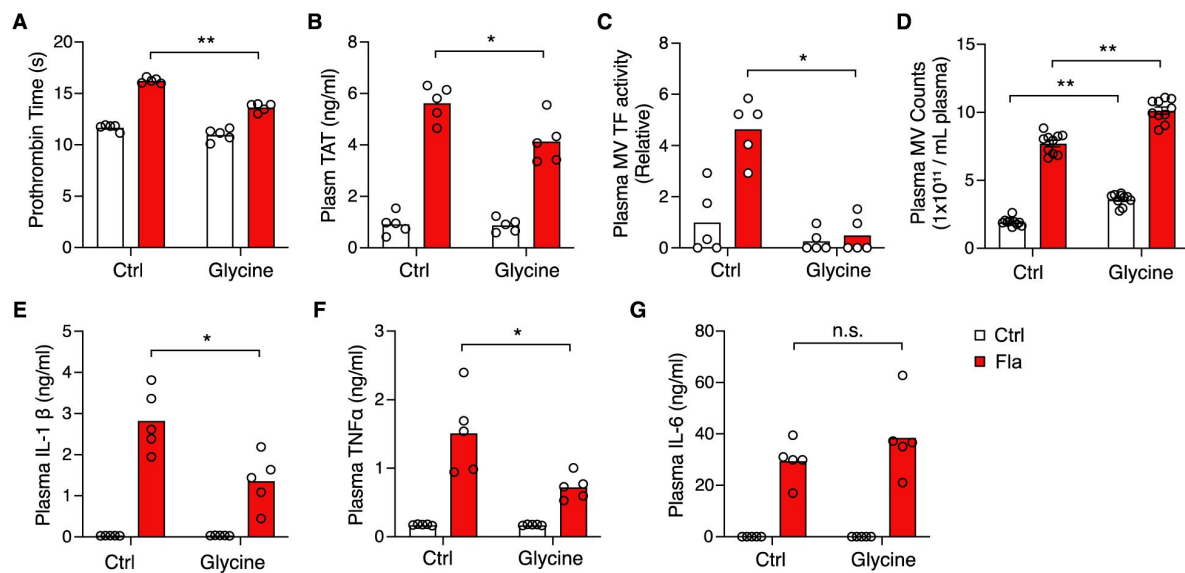


Figure 4.

Glycine inhibition of NINJ1 blocks pyroptosis-induced blood coagulation.

Mice were injected intravenously 50 μ l of 0.5 M glycine 2 hours before administrating Ctrl (saline) or Fla (500 ng LFn-Fla plus 3 μ g PA). Blood was collected 90 minutes afterwards. Prothrombin time (A), plasma TAT (B), plasma TF MV activity (C), plasma MV counts (D), and plasma cytokines (E-G) were measured. Circles represent individual mouse, with bars donating mean. * $p < 0.05$, ** $p < 0.01$, n.s. donates not significant (two-way ANOVA with Holm-Sidak multiple comparisons).

Methods

Mice

C57BL/6J, *Ninj1*^{+/+}, and *Ninj1*^{+/-} mice were housed in the University of Kentucky Animal Care Facility, following institutional and National Institutes of Health guidelines after approval by the Institutional Animal Care and Use Committee.

Recombinant protein purification

Recombinant proteins were expressed using LPS free *E. coli* strain ClearColi BL21(DE3) (Lucigen Corporation, Cat#60810) at 37 °C for 4 hours with 500 μM IPTG after OD600 reached 0.6-0.8. Bacteria were collected and lysed in 50 mM Tris-HCL and 300 mM NaCl. Proteins containing a His-tag were purified by affinity chromatography using HisPur Ni-NTA resin (Thermo Fisher Scientific, Cat#88222). Proteins were then eluted with 250 mM imidazole in 50 mM Tris-HCL and 300 mM NaCl, and subsequently dialyzed against PBS to remove imidazole. Proteins concentrations were determined by measuring A280 before sterile filtration.

In vivo Challenges

For flagellin challenge, purified PA and LFn-Fla in saline were administered via retro-orbital injection. Blood samples were collected at 90 minutes or 6 hours post-injection, followed by saline perfusion. Subsequently, tissue samples were harvested. For *E. coli* challenge, mice were injected intraperitoneally with *E. coli* saline solution. Blood samples were collected at 6 hours post-injection. For glycine *in vivo* experiment, mice were injected intravenously with glycine saline solution 2 hours before flagellin challenge. Blood samples were collected at 90 minutes after flagellin injection.

BMDM isolation and differentiation

Mouse femur and tibia from one leg are harvested and rinsed in ice-cold PBS, followed by a brief rinse in 70% ethanol for 10-15 seconds. Both ends of the bones are then cut open, and the bone marrow is flushed out using a 10 ml syringe with a 26-gauge needle. The marrow is passed through a 19-gauge needle once to disperse the cells. After filtering through a 70-μm cell strainer, the cells are collected by centrifugation at 250 g for 5 minutes at 4 °C, then suspended in two 150 mm petri dish, each containing 25 ml of L-cell conditioned medium (RPMI-1640 supplemented with 10% FBS, 2mM L-Glutamine, 10mM HEPES, 15% LCM, and penicillin/streptomycin). After 3 days, 15 mL of LCM medium is added to each dish cells. The cells typically reach full confluency by days 5-7.

BMDM culture and stimulation

BMDMs seeded into 12-well cell culture plate or 96-well cell culture plate at a density of 1×10^6 cells/mL of RPMI-1640 medium (Thermo Fisher Scientific, Cat# 61870036) containing 10% of Fetal Bovine Serum (FBS) (Thermo Fisher Scientific, Cat# A3160502). BMDMs were allowed to settle overnight and refreshed with Opti-MEM (Thermo Fisher Scientific, Cat# 51985034) before purified proteins were added. To measure cytotoxicity, cell viability and MVs, supernatant was collected 90 minutes after stimulation.

Cell viability and cytotoxicity

Cell viability of BMDMs was determined using CellTiter-Glo Luminescent Cell Viability Assay (Promega, Cat#G7572). Luminescence was recorded as an indicator of ATP concentrations in metabolically active cells. BMDMs cell cytotoxicity, as measured by LDH in the cell culture

supernatant, was determined using CytoTox 96 Non-Radioactive Cytotoxicity Assay (Promega, Cat#G1780) according to manufacturer's instruction.

Prothrombin Time (PT)

Blood samples were collected from ketamine/xylazine-anaesthetized mice by cardiac puncture with a 23-gauge needle attached to a syringe pre-filled with 3.8% trisodium citrate as anticoagulant (final ratio at 1:10). Blood was centrifuged at 2,000 g for 20 minutes at 4 °C to obtain plasma. Prothrombin time (PT) was determined with Thromboplastin-D (Pacific Hemostasis, Cat#100357) in a manual setting according to manufacturer's instruction, using CHRONO-LOG #367 plastic cuvette.

Plasma TAT

Plasma was collected as mentioned above in PT. Plasma TAT concentrations were determined using a mouse TAT ELISA kit (Abcam, Cat#ab137994) at 1:50 dilution according to manufacturer's instruction.

ELISA

IL-1 β , IL-6, and TNF α levels in culture supernatant and plasma were measured with ELISA kits from Thermo Fisher Scientific according to manufacturer's instruction. Plates were read on a Cytation 5 at 450 and 570 nm.

Tissue preparation and immunohistochemistry

Mice were perfused via both right and left ventricles with saline. Tissues were collected and embedded in paraffin, then sectioned at 5 μ m. Anti-fibrin antibody 59D8 (kindly provided by Dr. Hartmut Weiler at Medical College of Wisconsin and Dr. Rodney M. Camire at the University of Pennsylvania) at 4 mg/ml was used for staining fibrin deposition, with M.O.M.® (Mouse on Mouse) ImmPRESS® HRP (Peroxidase) Polymer Kit (Vector, Cat#MP-2400) from vector laboratories according to manufacturer's instruction for developing positive staining.

Fibrin extraction for immunoblot

Frozen tissues were homogenized in 20 volumes (mg: μ L) of T-PER tissue protein extraction reagent (Thermo Fisher Scientific, Cat#78510) containing cocktail inhibitor (Sigma, Cat#P8340). After centrifugation at 10,000 g for 10 minutes, supernatant was collected for β -actin detection. Pellets were then homogenized in 3 M urea and vortexed for 2 hours at 37 °C. After centrifugation at 14,000 g for 15 minutes, pellets were suspended in LDS sample buffer and vortexed at 65 °C for 30 minutes and ready for fibrin detection.

Fluorescent immunoblot

For detection of Caspase-1, IL-1 β , GSDMD, and NINJ1 by immunoblot, cells were washed with cold PBS and lysed with LDS sample buffer. Culture supernatant was precipitated with 1/10 volume of 2% sodium cholate and 1/10 volume of 100% trichloroacetic acid (TCA), and then dissolved in LDS sample buffer. TF was detected using anti-tissue factor (R&D, Cat#AF3178-SP) at 1:1000 dilution. Caspase-1 was detected using anti-Caspase-1 (Adipogen, Cat#AG-20B-0042-C100) at 1:1000 dilution. IL-1 β was detected using anti-IL-1 β (GeneTex, Cat#GTX74034) at 1:1000 dilution. GSDMD was detected using anti-GSDMD (Abcam, Cat#ab219800) at 1:1000 dilution. NINJ1 was detected using anti-NINJ1 (kindly provided by Genentech, Cat#Ninj1-rbIgG-25:10363). Tissue fibrin was detected using anti-fibrin (59D8) at 1 mg/ml. Images were acquired with LI-COR Odyssey Imager.

Isolation of MVs from mouse plasma

Plasma was collected as mentioned above in PT. Then 50 μ L of mouse plasma was diluted with 1 mL of HBSA (137 mM NaCl, 5.38 mM KCl, 5.55 mM glucose, 10 mM HEPES, 0.1% bovine serum albumin, pH 7.5). MVs were pelleted at 20,000 g for 60 minutes at 4 °C, washed once with 1 mL of HBSA and resuspended in 100 μ L HBSA.

Isolation of MVs from cell culture medium

Cell debris was removed from cell culture supernatant by centrifugation at 1,000 g for 10 minutes. MVs were then pelleted at 20,000 g for 60 minutes at 4 °C and resuspended in HBSA.

MV Counting

To count MVs collected from cell culture supernatant, cell debris was removed by centrifugation at 1000 g for 10 minutes. MVs in cell culture supernatant or plasma were analyzed directly without further centrifugation. All samples were diluted to optimal conditions (about 10^8 particles/mL) for analysis in PBS. Video recordings were made for 10 intervals at a length of 30 seconds each using NanoSight. Nanoparticle tracking analysis was performed to measure the number and size of the plasma and culture supernatant MVs.

Plasma MV TF Activity Assay

Plasma samples (25 μ L each) were incubated with the 1H1 anti-TF antibody (Genentech) at 100 mg/mL or rat IgG controls for 15 minutes at room temperature. Next, 25 μ L of HBSA containing 10 nM mouse FVIIa, 300 nM human FX and 10 mM CaCl₂ was added to the samples and incubated for 2 hours at 37 °C in a half area 96-well plate. Finally, 12.5 μ L of the FXa substrate RGR-XaChrom (4 mM, Enzyme Research Laboratories#100-03) was added and the mixture was incubated at 37 °C for 15 minutes. Absorbance at 405 nm was measured on Cytation 5. The relative TF activity was calculated with absorbance value after subtracting the TF-independent activity in the presence of TF blocking antibodies from the total activity in the presence of the IgG controls.

Statistical analysis

Data are represented as individual dots, with bar denoting mean. For multiple-group with two independent factors, two-way ANOVA with Holm-Sidak multiple comparisons was used for normally distributed variables. A p value <0.05 was considered significant. Statistical analyses were performed in Prism GraphPad Prism 9.

Supplementary Data

Figure S1.

NINJ1 protein abundance in different tissues.

Protein was extracted from fresh frozen tissues and detected by immunoblot.

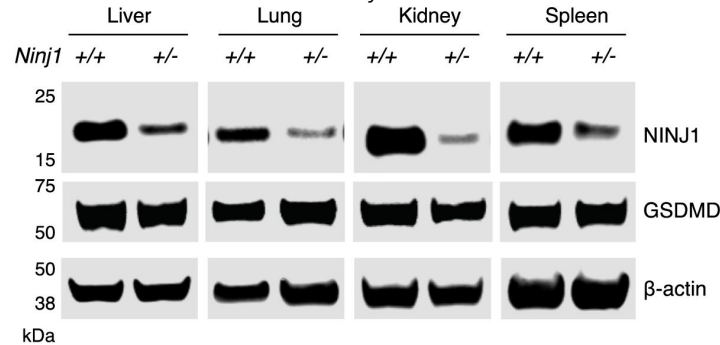
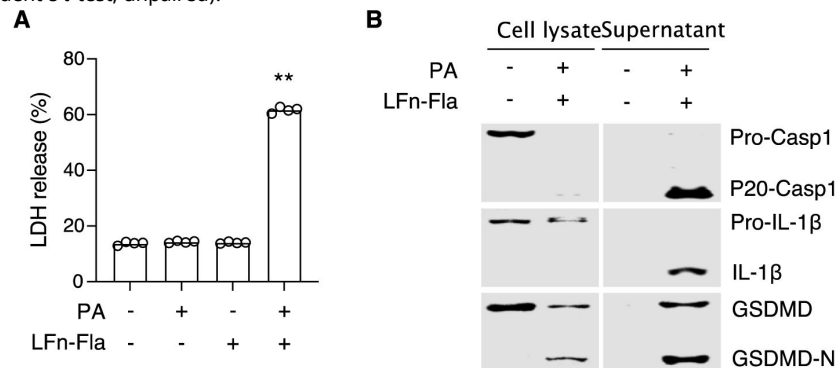


Figure S2.

Flagellin-induced inflammasome activation and pyroptosis. BMDMs from

Ninj1^{+/+} mice were incubated with LFn-Fla (1 μg/mL) and/or PA (1 μg /mL) for 90 minutes. **(A)** Plasma membrane rupture (PMR) was measured by LDH release. **(B)** Caspase-1, IL-1β, and GSDMD in the cell lysates and culture supernatant was detected by immunoblot. Circles represent individual mouse, with bars donating mean. ** *p* < 0.01 versus each of the other three groups (Student's *t*-test; unpaired).



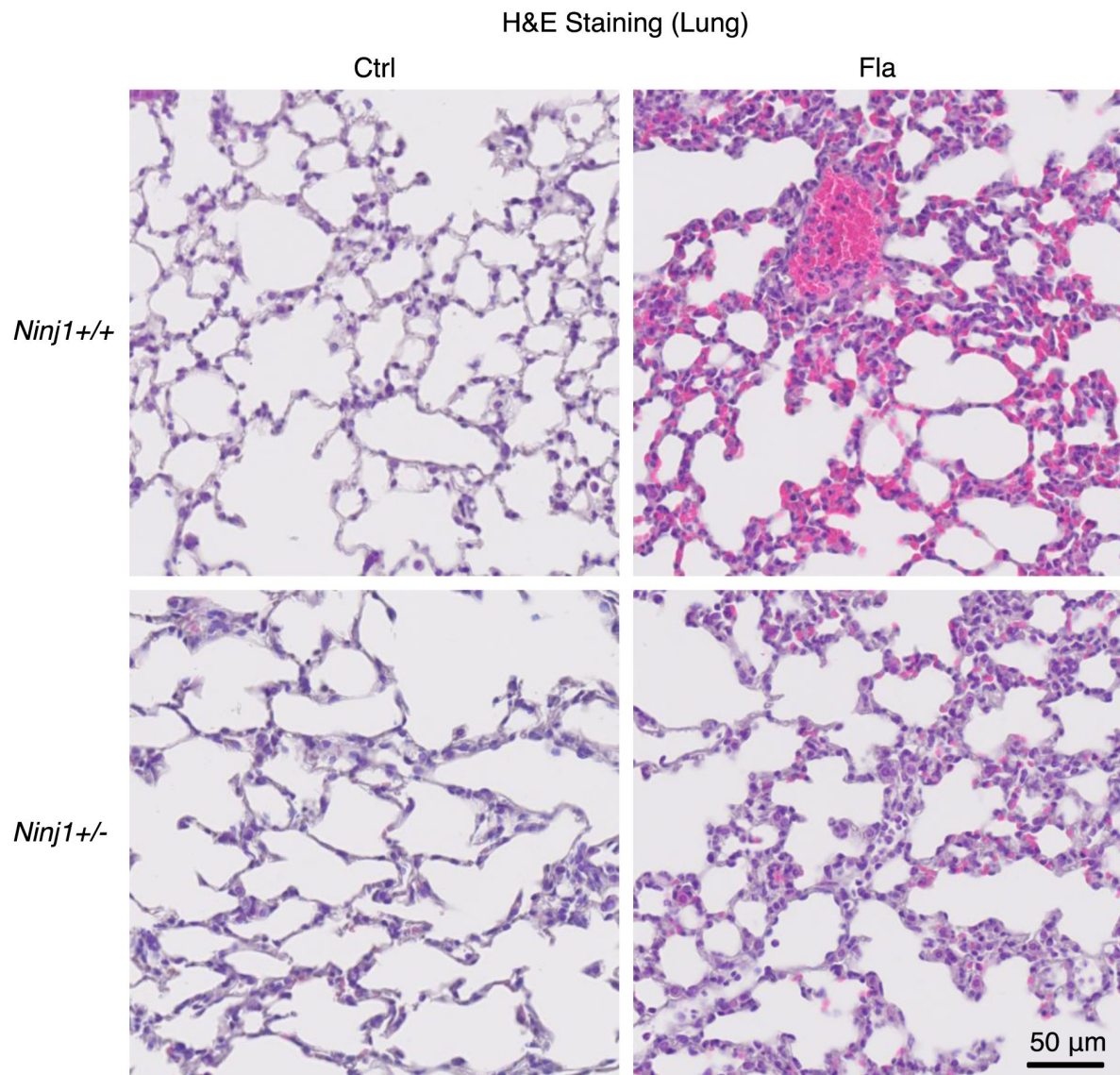


Figure S3.

Flagellin-induced pathology in *Ninj1*^{+/+} and *Ninj1*^{+/-}.

Mice were injected intravenously with Ctrl (saline) or Fla (500 ng LFn-Fla plus 3 μg PA). Tissues were collected 90 minutes after Ctrl or Fla injection. H&E staining was performed on saline-perfused and paraffin-embedded lung tissue sections.

Figure S4.

Flagellin-induced pyroptosis in *Ninj1*^{+/+} and *Ninj1*^{+/-} BMDMs.

BMDMs from *Ninj1*^{+/+} and *Ninj1*^{+/-} mice were incubated with Ctrl (PBS) or Fla (1 µg/mL LFn-Fla plus 1 µg/mL PA) for 90 minutes. LDH release (**A**) and ATP (**B**) were measured. Circles represent individual mouse, with bars donating mean. ** $p < 0.01$, n.s. donates not significant (two-way ANOVA with Holm-Sidak multiple comparisons).

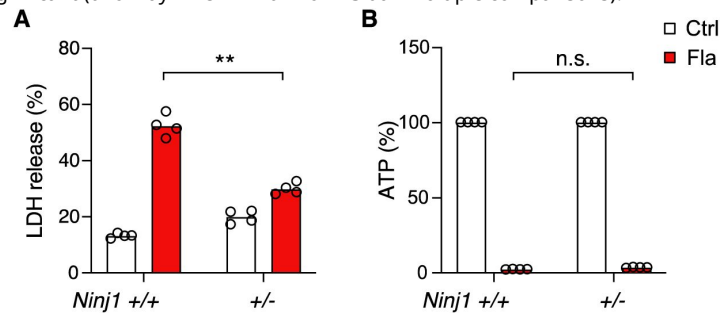
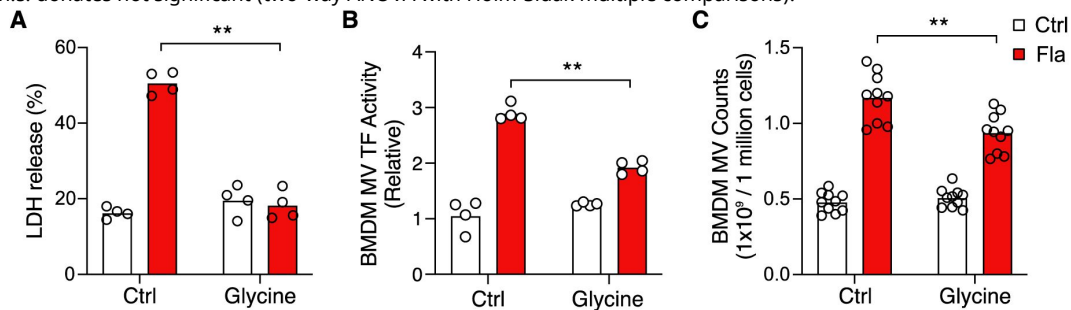


Figure S5.

Glycine treatment inhibits BMDM MV release.

(A-C) BMDMs from *Ninj1*^{+/+} mice were treated with 5 mM glycine for 30 minutes before incubation with Ctrl (PBS) or Fla (1 µg/mL LFn-Fla plus 1 µg/mL PA). Cell culture supernatant was collected after 90 minutes incubation. LDH release (**A**), BMDM MV TF activity (**B**), and MV counts (**C**) were measured. Circles represent individual mouse, with bars donating mean. ** $p < 0.01$, n.s. donates not significant (two-way ANOVA with Holm-Sidak multiple comparisons).



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Additional information

Author contributions

C.W. and J.C. designed and performed the experiments and wrote the manuscript. H.L., G.Z., Y.Z., L.Y., and M.M.S.S. assisted the experiments. J.P.W., Y.W. and Z.L. contributed to manuscript preparation. All authors discussed the results and commented on the manuscript.

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Reviewer #1 (Public review):

The authors demonstrated that NINJ1 promotes TF-positive MV release during pyroptosis and thereby triggers coagulation. Coagulation is one of the risk factors that can cause secondary complications in various inflammatory diseases, making it a highly important therapeutic target in clinical treatment. This paper effectively explains the connection between pyroptosis and MV release with Ninj1, which is a significant strength. It provides valuable insight into the potential of targeting Ninj1 as a therapeutic strategy.

Although the advances in this paper are valuable, several aspects need to be clarified. Some comments are discussed below.

(1) Since it is not Ninj1 directly regulating coagulation but rather the MV released by Ninj1 playing a role, the title should include that. The current title makes it seem like Ninj1 directly regulates inflammation and coagulation. It would be better to revise the title.

(2) Ninj1 is known to be an induced protein that is barely expressed in normal conditions. As you showed in "Fig1G" data, control samples showed no detection of Ninj1. However, in "Figure S1", all tissues (liver, lung, kidney and spleen) expressed Ninj1 protein. If the authors stimulated the mice with fla injection, it should be mentioned in the figure legend.

(3) In "Fig3A", the Ninj1 protein expression was increased in the control of BMDM +/- cell lysate rather than fla stimulation. However, in MV, Ninj1 was not detected at all in +/- control but was only observed with Fla injection. The authors need to provide an explanation for this observation. Additionally, looking at the MV β -actin lane, the band thicknesses appear to be very different between groups. It seems necessary to equalize the protein amounts. If that is difficult, at least between the +/- and +/- controls.

(4) Since the authors focused Ninj1-dependent microvesicle (MV) release, they need to show MV characterizations (EM, NTA, Western for MV markers, etc...).

(5) To clarify whether Ninj1-dependent MV induces coagulation, the authors need to determine whether platelet aggregation is reduced with isolated +/- MVs compared to +/- MVs.

(6) Even with the authors well established experiments with haploid mice, it is a critical limitation of this paper. To improve the quality of this paper, the authors should consider confirming the findings using mouse macrophage cell lines, such as generating Ninj1-/- Raw264.7 cell lines, to examine the homozygous effect.

(7) There was a paper reported in 2023 (Zhou, X. et al., NINJ1 Regulates Platelet Activation and PANoptosis in Septic Disseminated Intravascular Coagulation. *Int. J. Mol. Sci.* 2023) that revealed the relationship between Ninj1 and coagulation. According to this paper, inhibition of Ninj1 in platelets prevents pyroptosis, leading to reduced platelet activation and, consequently, the suppression of thrombosis. How about the activation of platelets in Ninj1 +/- mice? The author should add this paper in the reference section and discuss the platelet functions in their mice.

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Reviewer #2 (Public review):

Summary:

The authors main goal is to understand the mechanism by which pyroptosis (through the formation of Gasdermin D (GSDMD) pores in the plasma membrane) contributes to increased release of procoagulant Tissue Factor-containing microvesicles (MV). Their previous data demonstrate that GSDMD is critical for the release of MV that contains Tissue Factor (TF), thus making a link between pyroptosis and hypercoagulation. Given the recent identification of NINJ1 being responsible for plasma membrane rupture (Kayagaki et al. *Nature* 2011), the authors wanted to determine if NINJ1 is responsible for TF-containing MV release. Given the constitutive ninj1 KO mouse leads to partial embryonic lethality, the authors decide to use a heterozygous ninj1 KO mouse (ninj1 +/-), and demonstrate that Ninj1 plays a role in release of TF-containing MV.

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #1 (Public Review):

The authors demonstrated that NINJ1 promotes TF-positive MV release during pyroptosis and thereby triggers coagulation. Coagulation is one of the risk factors that can cause secondary complications in various inflammatory diseases, making it a highly important therapeutic target in clinical treatment. This paper effectively explains the connection between pyroptosis and MV release with Ninj1, which is a significant strength. It provides valuable insight into the potential of targeting Ninj1 as a therapeutic strategy.

Although the advances in this paper are valuable, several aspects need to be clarified. Some comments are discussed below.

(1) Since it is not Ninj1 directly regulating coagulation but rather the MV released by Ninj1 playing a role, the title should include that. The current title makes it seem like Ninj1 directly regulates inflammation and coagulation. It would be better to revise the title.

Thanks for the thoughtful comments. We show that the release of procoagulant MVs by plasma membrane rupture (PMR) is a critical step in the activation of coagulation. In addition, the release of cytokines and danger molecules by PMR may also contribute to coagulation. In choosing the title, we are trying to emphasize NINJ1-dependent PMR as a common trigger for these biological processes.

(2) Ninj1 is known to be an induced protein that is barely expressed in normal conditions. As you showed in "Fig1G" data, control samples showed no detection of Ninj1. However, in "Figure S1", all tissues (liver, lung, kidney and spleen) expressed Ninj1 protein. If the authors stimulated the mice with fla injection, it should be mentioned in the figure legend.

We respectfully disagree with the comment that “Ninj1 is known to be an induced protein that is barely expressed in normal conditions”. NINJ1 protein is abundantly expressed (without induction) in tissues including liver, lung, kidney, and spleen, as shown in Fig S1. Consistently, other groups have shown abundant NINJ1 expression at baseline in tissues and cells such as liver (Kayagaki *et.al.* Nature 2023) and BMDM (Kayagaki *et.al.* Nature 2021; Borges *et.al.* eLife 2023). Fig 1G shows fibrin deposition as an indicator of coagulation, not NINJ1 protein.

(3) In "Fig3A", the Ninj1 protein expression was increased in the control of BMDM +/- cell lysate rather than fla stimulation. However, in MV, Ninj1 was not detected at all in +/- control but was only observed with Fla injection. The authors need to provide an explanation for this observation. Additionally, looking at the MV β -actin lane, the band thicknesses appear to be very different between groups. It seems necessary to equalize the protein amounts. If that is difficult, at least between the +/+ and +/- controls.

Thanks for the valuable comments. In Fla-stimulated Ninj1^{+/-} BMDMs, most of the NINJ1 is released in MVs, therefore, not in the cell lysate, as shown in Fig 3A. The difference in beta-actin band intensity correlated with MV numbers shown in Fig 3B. We ensure consistency by using the same number of cells.

(4) Since the authors focused *Ninj1*-dependent microvesicle (MV) release, they need to show MV characterizations (EM, NTA, Western for MV markers, etc...).

Thanks for the suggestion. We now add NTA analysis of MV for BMDMs in Fig S4C.

(5) To clarify whether *Ninj1*-dependent MV induces coagulation, the authors need to determine whether platelet aggregation is reduced with isolated +/- MVs compared to +/- MVs.

Thanks for the suggestion. We agree that platelet aggregation is closely linked to blood coagulation but would argue that one does not directly cause the other. While it would be interesting to examine whether MVs induce platelet aggregation, we hope the reviewer would agree that the outcome of this experiment would neither significantly support nor challenge our statement that *NINJ1*-dependent PMR promotes coagulation.

(6) Even with the authors well established experiments with haploid mice, it is a critical limitation of this paper. To improve the quality of this paper, the authors should consider confirming the findings using mouse macrophage cell lines, such as generating *Ninj1*^{-/-} Raw264.7 cell lines, to examine the homozygous effect.

Thanks for the valuable comments. We acknowledge the limitation of using haploid mice in this study. However, our data provides strong evidence supporting the role of *NINJ1*-dependent plasma membrane rupture in blood coagulation using primary macrophages.

(7) There was a paper reported in 2023 (Zhou, X. et al., *NINJ1 Regulates Platelet Activation and PANoptosis in Septic Disseminated Intravascular Coagulation*. *Int. J. Mol. Sci.* 2023) that revealed the relationship between *Ninj1* and coagulation. According to this paper, inhibition of *Ninj1* in platelets prevents pyroptosis, leading to reduced platelet activation and, consequently, the suppression of thrombosis. How about the activation of platelets in *Ninj1* +/- mice? The author should add this paper in the reference section and discuss the platelet functions in their mice.

Thanks for the valuable comments. We examine PT time, plasma TAT, and tissue fibrin deposition as direct evidence of blood coagulation in this manuscript. We acknowledge that platelets play a key role in thrombosis; however, we hope the reviewer would agree that tissue factor-induced blood coagulation and platelet aggregation are linked yet distinct processes. Therefore, the role of *NINJ1* in platelet aggregation falls beyond the scope of this manuscript.

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Referring to previous research findings, the authors explain the connection between *NINJ1* and MVs. Additional experiments and clarifications will strengthen the conclusions of this study.

Below are some comments I feel could strengthen the manuscript:

(1) The authors mentioned their choice of using heterozygous *NINJ1*^{+/-} mice on page 4, because of lethality and hydrocephalus. Nonetheless, there is a substantial number of

references that use homozygous NINJ1^{-/-} mice. Could there be any other specific reasons for using heterozygous mice in this study?

Thanks for the thoughtful comments. We are aware that a few homozygous NINJ1^{-/-} mouse strains were used in several publications by different groups, including Drs. Kayagaki and Dixit (Genentech), from whom we obtained the heterozygous NINJ1^{+/-} breeders. We do not have experience with the homozygous NINJ1^{-/-} mice used by other groups. It's reasonable to assume that homozygous NINJ1^{-/-}, if healthy, would have even stronger protection against coagulopathy than heterozygous NINJ1^{+/-}. The only reason for not using homozygous mice in this study is that a majority of our homozygous NINJ1^{-/-} develops hydrocephalus around weaning and these mice are required to be euthanized by the rules of our DLAR facility. Although our homozygous NINJ1^{-/-} mice develop hydrocephalus (the same reported by Drs. Kayagaki and Dixit, PMID: 37196676, PMCID: PMC10307625), heterozygous NINJ1^{+/-} mice remain healthy.

(2) Figure S2 clearly shows the method of pyroptosis induction by flagellin. It is also necessary as a prerequisite for this paper to show the changes in flagellin-induced pyroptosis in heterozygous NINJ1^{+/-} mice.

Thanks for the valuable suggestions. We agree that a plasma LDH measurement as an indicator of pyroptosis *in vivo* would add to the manuscript. Therefore, we have made several attempts to measure plasma LDH in flagellin-challenged WT and NINJ1^{+/-} mice using CytoTox96 Non-Radioactive Cytotoxicity Assay (a Promega kit commonly used for LDH, Promega#G1780). Flagellin-challenged WT and NINJ1^{+/-} mice develops hemolysis, which renders plasma red. Because plasma coloring interferes with the assay, we could not get a meaningful reading to make an accurate comparison. We also tried LHD-Glo Cytotoxicity Assay (Luciferase based, Promega#J2380) with no luck on both plasma and serum. We hope the reviewer would agree that reduced plasma MV count (Fig 3C) would serve as an alternative indicator for reduced pyroptosis.

(3) IL-1 β levels controlled by GSDMD were not affected by NINJ1 expression according to previous studies (Ref 37, 29, Nature volume 618, pages 1065-1071 (2023)). GSDMD also plays an important role in TF release in pyroptosis. Are GSDMD levels not altered in heterozygous NINJ1^{+/-} mice?

Thanks for raising these great points. It's been reported that IL-1 β secretion in cell culture supernatant were not affected by NINJ1 deficiency or inhibition when BMDMs were stimulated by LPS (Ref 29, 37, now Ref 29, 35) or nigericin (Ref 29). As GSDMD pore has been shown to facilitate the release of mature IL-1 β , these *in vitro* observations are reasonable given that NINJ1-mediated PMR is a later event than GSDMD pore-forming. However, we observed that plasma IL-1 β (also TNF α and IL-6) in NINJ1^{+/-} mice were significantly lower. There are a few differences in the experimental condition that might contribute to the discrepancy: 1, there was no priming in our *in vivo* experiment, while priming in BMDMs were performed in both *in vitro* observations before stimulating with LPS or nigericin; 2, the flagellin in our study engages different inflammasome than either LPS or nigericin. Priming might change the expression and dynamics of IL-1 β . More importantly, there might be unrecognized mechanisms in IL-1 β secretion *in vivo*. We now add discussion on this in the main text.

We examined GSDMD protein levels in liver, lung, kidney, and spleen from WT and NINJ1^{+/-} mice by Western blotting. The data is now presented in the updated Fig S1, we did not observe apparent difference in GSDMD expression between the two genotypes.

(4) In Fig 1 F, the authors used a fibrin-specific monoclonal antibody for staining fibrin, but it's not clearly defined. There may be some problem with the quality of antibody or technical issues. Considering this, exploring alternative methods to visualize fibrin might be beneficial. Fibrin is an acidophil material, so attempting H&E staining or Movat's pentachrome staining might help for identify fibrin areas.

Thanks for the valuable suggestions. The fibrin-specific monoclonal antibody in our study is mouse anti-fibrin monoclonal antibody (59D8). This antibody has been shown to bind to fibrin even in the presence of human fibrinogen at the concentration found in plasma [Hui et al. (1983). Science. 222 (4628); 1129-1132]. We apologize that we did not cite the reference in our initial submission. We obtained this antibody from Dr. Hartmut Weiler at Medical College of Wisconsin and Dr. Rodney M. Camire at the University of Pennsylvania, who were acknowledged in our initial submission.

We performed H&E staining on serial sections of the same tissues for Figure 1F. The data is now presented as Fig S3.

Reviewer #2 (Public Review):

Summary:

The author's main goal is to understand the mechanism by which pyroptosis (through the formation of Gasdermin D (GSDMD) pores in the plasma membrane) contributes to increased release of procoagulant Tissue Factor-containing microvesicles (MV). Their previous data demonstrate that GSDMD is critical for the release of MV that contains Tissue Factor (TF), thus making a link between pyroptosis and hypercoagulation. Given the recent identification of NINJ1 being responsible for plasma membrane rupture (Kayagaki et al. Nature 2011), the authors wanted to determine if NINJ1 is responsible for TF-containing MV release. Given the constitutive ninj1 KO mouse leads to partial embryonic lethality, the authors decided to use a heterozygous ninj1 KO mouse (ninj1+/-). While the data are well controlled, there is limited understanding of the mechanism of action. Also, given that the GSDMD pores have an ~18 nm inner diameter enough to release IL-1 β , while larger molecules like LDH (140 kDa) and other DAMPs require plasma membrane rupture (likely mediated by NINJ1), it s not unexpected that large MVs require NINJ1-mediated plasma cell rupture.

Strengths:

The authors convincingly demonstrate that ninj1 haploinsufficiency leads to decreased prothrombin time, plasma TAT and plasma cytokines 90 minutes post-treatment in mice, which leads to partial protection from lethality.

Weaknesses:

- In the abstract, the authors say "...cytokines and protected against blood coagulation and lethality triggered by bacterial flagellin". This conclusion is not substantiated by the data, as you still see 70% mortality at 24 hours in the ninj1+/- mice.

Thanks for the thoughtful comments. We corrected the text to “partially protected against blood coagulation and lethality triggered by bacterial flagellin”.

- The previous publication by the authors (Wu et al. Immunity 2019) clearly shows that GSDMDdependent pyroptosis is required for inflammasome-induced coagulation and mouse lethality. However, as it is not possible for the authors to use the homozygous ninj1 KO mouse due to partial embryonic lethality, it becomes challenging to compare

these two studies and the contributions of GSDMD vs. NINJ1. Comparing the contributions of GSDMD and NINJ1 in human blood-derived monocytes/macrophages where you can delete both genes and assess their relevant contributions to TF-containing MV release within the same background would be crucial in comparing how much contribution NINJ1 has versus what has been published for GSDMD? This would help support the in vivo findings and further corroborate the proposed conclusions made in this manuscript.

Thanks for the valuable question. We have shown that plasma MV TF activity was reduced in both GSDMD deficient mice (Ref 23) and *Ninj1*^{+/-} mice (present manuscript). Given that TF is a plasma membrane protein, MV TF most likely comes from ruptured plasma membrane. In flagellin-induced pyroptosis, both GSDMD and NINJ1 deficiency equally blocked LDH release (plasma membrane rupture) in BMDMs (Ref 29). Further, in pyroptosis glycine acts downstream of GSDMD pore formation for its effect against NINJ1 activation (Ref 35). Therefore, GSDMD pore-forming should be upstream of NINJ1 activation in pyroptosis (which may not be the case in other forms of cell death) and there are likely equal effects of GSDMD and NINJ1 on MV release in flagellin-induced pyroptosis. As the reviewer suggested, experiments using human blood-derived monocytes/macrophages will enable a direct comparison to determine the relative contribution. However, this approach presents a few technical difficulties: it's not easy to manipulate gene expression on primary human monocytes/macrophages (in our experience); variable efficiency in gene manipulation of GSDMD and NINJ1 will complicate the comparison. I hope the reviewer would agree that a direct comparison between GSDMD and NINJ1 is not required to support our conclusion that NINJ1-dependent membrane rupture is involved in inflammasome-pyroptosis induced coagulation and inflammation.

*- What are the levels of plasma TAT, PT, and inflammatory cytokines if you collect plasma after 90 minutes? Given the majority (~70%) of the *ninj1*^{+/-} mice are dead by 24 hours, it is imperative to determine whether the 90-minute timeframe data (in Fig 1A-G) is also representative of later time points. The question is whether *ninj1*^{+/-} just delays the increases in prothrombin time, plasma TAT, and plasma cytokines.*

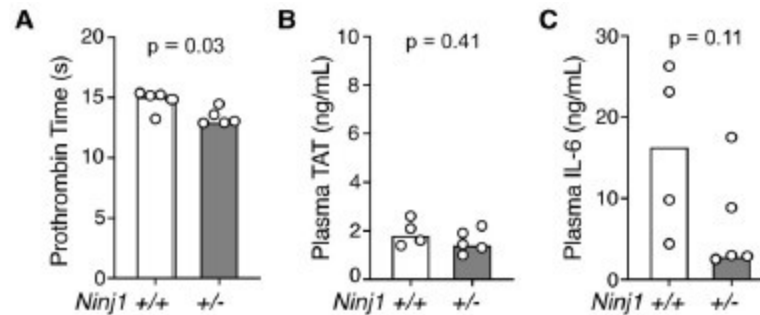
Thank for the valuable question. The time point (90 min) was chosen based on our *in vitro* observation that flagellin-induced pyroptosis in BMDMs largely occurs within 60-90 min.

Because our focus on the primary effect of flagellin *in vivo*, potential secondary effects at later points may complicate the results and are hard to interpret. As the reviewer suggested, we have measured plasma PT, TAT at 6 hours post-flagellin challenge. The significant difference in PT sustained between *Ninj1*^{+/+} and *Ninj1*^{+/-} (Fig A), suggesting coagulation proteins remained more depleted in *Ninj1*^{+/+} mice than in *Ninj1*^{+/-} mice. However, plasma TAT levels were diminished to baseline level (refer to Fig 1B in main text) in both groups and showed no significant difference between groups (Fig B), which could be explained by the short half-life (less than 30 min) in the blood. Since flagellin challenge is a one-time hit, there might not a second episode of coagulation after the 90-minute time point, at least not triggered by flagellin, supported by the plasma TAT levels at 6 hours. We now comment on this limitation at the end of the main text.

Based on our previous studies, plasma IL-1 β and TNF α peaked at early time point and diminished over time, but plasma IL-6 levels maintained. As shown below, plasma IL-6 appeared higher in *Ninj1*^{+/+} compared with *Ninj1*^{+/-}, but not statistically significant (partly because one missing sample, n = 4 not 5, in *Ninj1*^{+/+} group decreased the statistical power of detecting a difference).

Author response image 1.

Mice were injected with Fla (500 ng lFn-Fla plug3 ugPA). Blood was collected 6 hours after Fla injection. Prothrombin time (A), plasma TAT (B), and plasma IL-6 (C) were measured. Mann-Whitney test were performed.



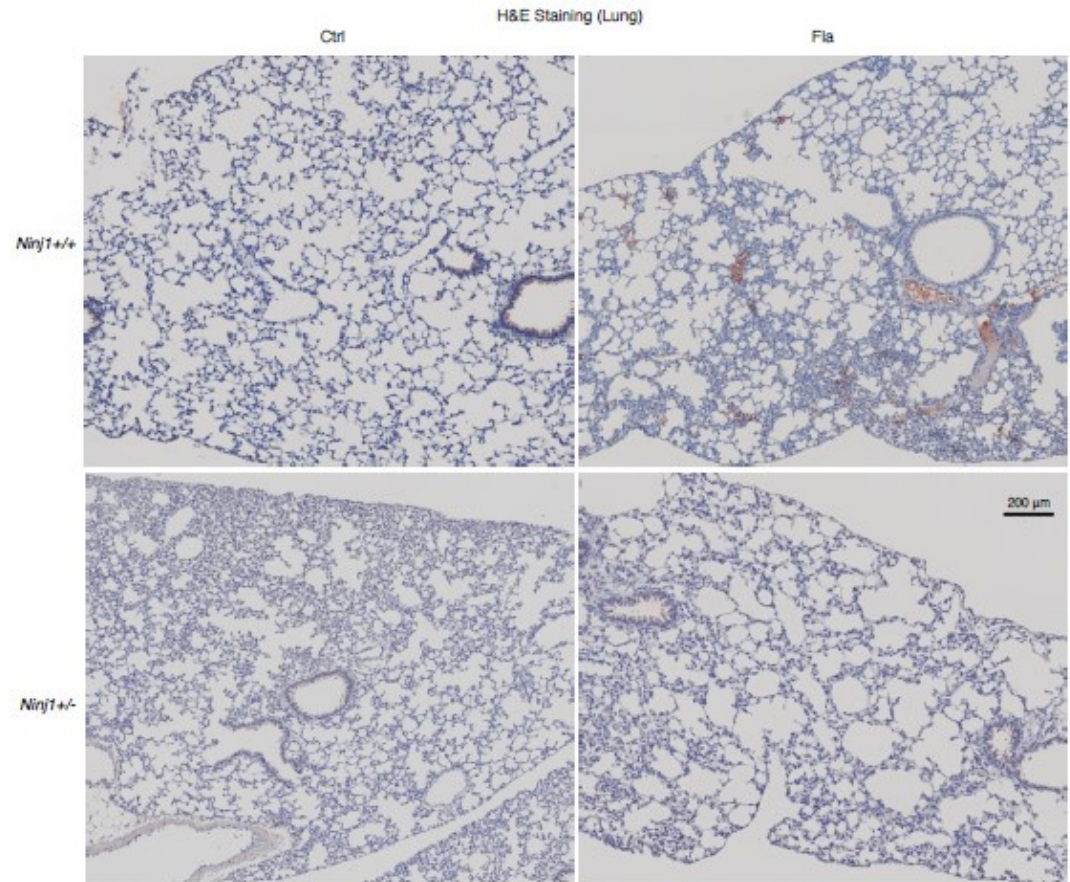
Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

- Fig 1F: are there lower magnification images that capture the fibrin deposition? The IHC data seems at odds with the WB data in Fig. 1G where there is still significant fibrin detected in the heterozygous lungs and liver. Quantitating the Fig. 1G Western blot would also be helpful.

IHC surveys a thin layer of tissue section while WB surveys a piece of tissue, therefore fibrin deposition may be missing from IHC and but found in WB. That is why we used two methods. Below we provide lower mag images of fibrin deposition (about 2 x 1.6 mm area).

Author response image 2.



- Fig1H - lethality study uses 5x dose of Fla used in earlier studies. In the lethality data where there is a delay in *ninj1*^{+/-} mortality, are the parameters (prothrombin time, plasma TAT, and plasma cytokines) measured at 90 minutes different between WT and *ninj*^{+/-} mice? This would be critical to confirm that this is not merely due to a delayed release of TF-containing MVs.

We used 5x lower dose of Fla in coagulation study than lethality study because it's not as easy to draw blood from septic mouse with higher dose of flagellin. We need to terminate the mice to collect blood for plasma measurement and therefore the parameters were not measured for mice in lethality study.

- What is the effect of *ninj*^{+/-} on *E. coli*-induced lethality in mice? How do these data compare to *E. coli* infection of *GSDMD*^{-/-} mice?

We did not examine the effect of *Ninj1*^{+/-} on *E. coli*-induced lethality. After the initial submission of our manuscript, we have focused on *Ninj1* flox/flox mice instead of *Ninj1*^{+/-} for *NINJ1* deficiency. We are using induced global *Ninj1* deficient mice for polymicrobial infection-induced lethality in our new studies.

- Fig 2 - in the *E. coli* model, the prothrombin time, plasma TAT, and plasma cytokines are measured 6 hours post-infection. How were these time points chosen? Did the authors measure prothrombin time, plasma TAT, and plasma cytokines at different time points?

The *in vivo* time point for flagellin and *E.coli* were chosen based on our *in vitro* observation of the timelines on BMDM pyroptosis induced by flagellin and bacteria. This disparity probably arises from distinct dynamics between purified protein and bacterial infections. Purified proteins can swiftly translocate into cells and take effect immediately after injection. Conversely, during bacterial infection, macrophages engulf and digest the bacteria to expose their antigens. Subsequently, these antigens initiate further effects, a process that takes some time to unfold.

Our focus is on the primary effect of flagellin *in vivo*, potential secondary effects at later points may complicate the results and are hard to interpret. As the reviewer suggested, we have measured plasma PT, TAT at 6 hours post-flagellin challenge. The significant difference in PT sustained between Ninj1^{+/+} and Ninj1^{+/-} (Fig A), suggesting coagulation proteins remained more depleted in Ninj1^{+/+} mice than in Ninj1^{+/-} mice. However, plasma TAT levels were diminished to baseline level (refer to Fig 1B in main text) in both groups and showed no significant difference between groups (Fig B), which could be explained by the short half-life (less than 30 min) in the blood. Since flagellin challenge is a one-time hit, there might not a second episode of coagulation after the 90-minute time point, at least not triggered by flagellin, supported by the plasma TAT levels at 6 hours. We now comment on this limitation at the end of the main text.

Based on our previous studies, plasma IL-1 β and TNF α peaked at early time point and diminished over time, but plasma IL-6 levels maintained. As shown below, plasma IL-6 appeared higher in Ninj1^{+/+} compared with Ninj1^{+/-}, but not statistically significant (partly because one missing sample, n = 4 not 5, in Ninj1^{+/+} group decreased the statistical power of detecting a difference).

- Fig 3 - the sequence of figure panels listed in the legend needs to be corrected. Fig 3A requires quantitation of NINJ1 levels compared to beta-actin. Fig 3C - needs a control for equal MV loading.

Thanks for the recommendations. The figure sequence has been corrected. There remain no common markers or loading controls for MV, so we use equal plasma volume for loading control.

Additional comments:

(1) In Fig 3A, the size of NINJ1 appears to be increased in the NINJ^{+/-} group.

This discrepancy is likely attributed to a technical issue when running the protein gel and protein transfer, which makes the image tilt to one side.

(2) Describe the method of BMDM isolation.

Thanks for the recommendations. We now include the method of BMDM isolation. In brief, mouse femur and tibia from one leg are harvested and rinsed in ice-cold PBS, followed by a brief rinse in 70% ethanol for 10-15 seconds. Both ends of the bones are then cut open, and the bone marrow is flushed out using a 10 ml syringe with a 26-gauge needle. The marrow is passed through a 19-gauge needle once to disperse the cells. After filtering through a 70- μ m cell strainer, the cells are collected by centrifugation at 250 g for 5 minutes at 4 °C, then suspended in two 150 mm petri dish, each containing 25 ml of L-cell conditioned medium (RPMI-1640 supplemented with 10% FBS, 2mM L-Glutamine, 10mM HEPES, 15% LCM, and penicillin/streptomycin). After 3 days, 15 mL of LCM medium is added to each dish cells. The cells typically reach full confluency by days 5-7.

(3) According to this method, BMDMs are seeded without any M-CSF or L929-cell conditioned medium. How many macrophages survive under this condition?

BMDMs are cultured and differentiated in medium supplemented with 15% L929-cell conditioned medium. For the experiment, the cells were seeded in Opti-MEM medium (Thermo Fisher Scientific, Cat# 51985034) without M-CSF or L929-cell conditioned medium. BMDMs can survive under this condition, as evidenced by low LDH and high ATP measurement (Fig S5).

Reviewer #2 (Recommendations For The Authors):

- There is significant information missing in the methods and this makes it unclear how to interpret how some of the experiments were performed. For example, there is no detailed description or references in the methods on how the *in vivo* experiments were performed. The methods section needs significantly more details so that any reader is able to follow the protocols in this manuscript. References to previous work should also be included as needed.

Thanks for the recommendations. We had some of the details in the figure legend. We now add details in the methods for better interpretation of our data.

- Line numbers in the manuscript would be helpful when resubmitting the manuscript so that the reviewer can easily point to the main text when making comments.

Thanks for the recommendations. We now add line numbers in the manuscript.

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